Lecture 19: Calculus with Parametric Curves

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Differentiate parametric equations to find the first derivative and second derivative.
- ☐ Determine the slope of the tangent line at a given point.
- ☐ Find the area under a parametric curve when bounded by the x-axis or y-axis.
- ☐ Determine the length of a curve defined parametrically.

Derivative of a Parametrized Curve

A parametrized curve $x = f(t)$ and $y = g(t)$ is differentiable at t if f and g are each differentiable at t.

If y is differentiable function of x, then by the chain rule,

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}. \quad \Rightarrow \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

denominator cannot equal to 0.

If all three derivatives exist and $dx/dt \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

If y is twice-differentiable function of x, then at any point where $dx/dt \neq 0$ and $y' = dy/dx$,

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy'}{dx}\right) = \frac{dy'/dt}{dx/dt}.$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}.$$

Derivative of a Parametrized Curve

Example: Suppose that a curve $\mathcal{C}$ is defined as

$$x = t^2 - 1, \quad y = \frac{3}{t}.$$

(a) Find a Cartesian relation between x and y.
(b) Which point on the curve corresponds to $t = 6$?
(c) Find $\frac{dy}{dx}$ at the point $(8, 1)$ by using the parametric differentiation.

(a) $y = \frac{3}{t} \Rightarrow t = \frac{3}{y}$

So, $x = t^2 - 1 \Rightarrow x = \left(\frac{3}{y}\right)^2 - 1 \Rightarrow y^2 = \frac{9}{x+1}$

(b) At $t=6$, $x = (6)^2 - 1 = 35$

$y = \frac{3}{6} = \frac{1}{2}$

So, $(x, y) = \left(35, \frac{1}{2}\right)$.

(c) At $(8,1)$, $y = \frac{3}{t} = 1 \Rightarrow t = 3$

(Check, $t = 3 \Rightarrow x = (3)^2 - 1 = 8$ ✓)

So at $(8,1)$, $t = 3$.

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\left(-\frac{3}{t^2}\right)}{2t} = \frac{-\frac{3}{3^2}}{2(3)} = -\frac{1}{18}$$

Check: $y^2 = \frac{9}{x+1}$

$$\Rightarrow 2y\frac{dy}{dx} = \frac{-9}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-9}{2y(x+1)^2}$$

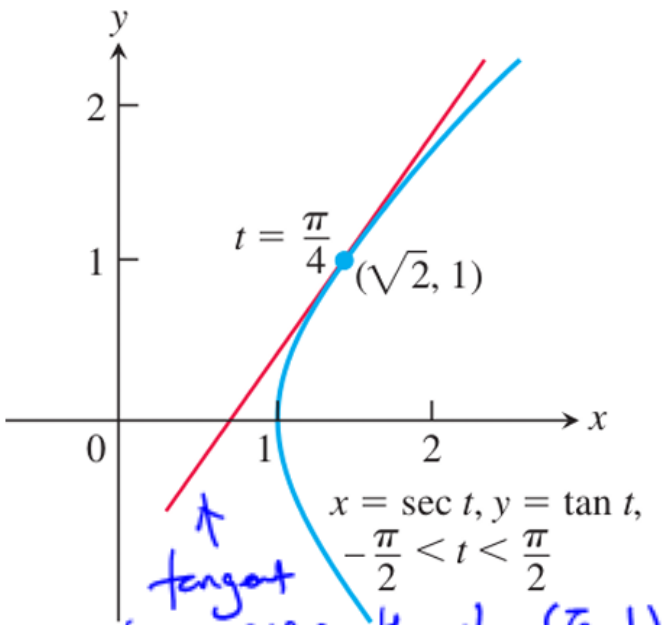
At $(8,1)$, $\frac{dy}{dx} = \frac{-9}{2(1)(8+1)^2} = -\frac{1}{18}$.

Tangent Lines and Areas

Example: Find the tangent line to the curve

$$x = \sec t, \quad y = \tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2},$$

at the point $(\sqrt{2}, 1)$, where $t = \frac{\pi}{4}$.



tangent line passing through $(\sqrt{2}, 1)$ with gradient $\sqrt{2}$.

$$\frac{dx}{dt} = \sec t \tan t.$$

$$\frac{dy}{dt} = \sec^2 t$$

So the slope of the curve at t is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sec^2 t}{\sec t \tan t} = \frac{\sec t}{\tan t}$$

At $t = \pi/4$, $\frac{dy}{dx} = \frac{\sec \pi/4}{\tan \pi/4} = \frac{2}{\sqrt{2}} = \sqrt{2}$

Eqn of the tangent line

$$(y-1) = \sqrt{2}(x-\sqrt{2}) \Rightarrow y = \sqrt{2}x - 1$$

Tangent Lines and Areas

Example: Find $\frac{d^2y}{dx^2}$ as a function of t if $x = t - t^2$ and $y = t - t^3$.

$$\frac{dx}{dt} = 1 - 2t$$

$$\frac{dy}{dt} = 1 - 3t^2$$

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1-3t^2}{1-2t}$$

$$\frac{dy'}{dt} = \frac{-6t(1-2t) - (-2)(1-3t^2)}{(1-2t)^2} = \frac{-6t + 12t^2 + 2 - 6t^2}{(1-2t)^2}$$

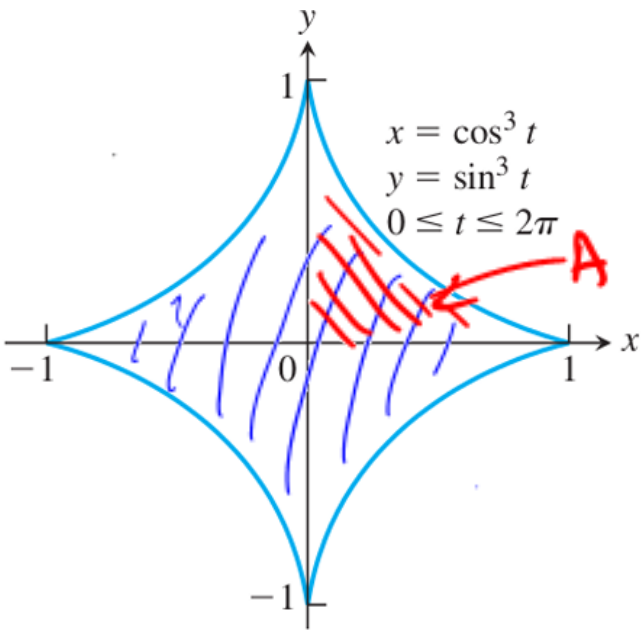
$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{\frac{6t^2 - 6t + 2}{(1-2t)^2}}{(1-2t)} = \frac{6t^2 - 6t + 2}{(1-2t)^2}$$

$$= \frac{6t^2 - 6t + 2}{(1-2t)^3}$$

Tangent Lines and Areas

Example: Find the area enclosed by the astroid

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \le t \le 2\pi.$$



By symmetry, Area enclosed = $4A$

$$A = \int_0^1 y\,dx$$

$y = \sin^3 t$

$x = \cos^3 t$

$dx = 3\cos^2 t(-\sin t)\,dt$

$x = 0 \Rightarrow t = \pi/2$, $x = 1 \Rightarrow t = 0$

$$= \int_{\pi/2}^{0} \sin^3 t \, 3\cos^2 t(-\sin t)\,dt$$

$$= \int_0^{\pi/2} 3\sin^4 t \cos^2 t \,dt.$$

refer to ch 8.

$$= 3\int_0^{\pi/2} \left(\frac{1-\cos 2t}{2}\right)^2 \left(\frac{1+\cos 2t}{2}\right) dt$$

$$= \frac{3}{8}\int_0^{\pi/2} (1 - 2\cos 2t + \cos^2 2t)(1 + \cos 2t)\,dt$$

$$= \frac{3}{8}\int_0^{\pi/2} 1 + \cos 2t - 2\cos 2t - 2\cos^2 2t + \cos^2 2t + \cos^3 2t \,dt$$

$$= \frac{3}{8}\int_0^{\pi/2} 1 - \cos 2t - \cos^2 2t + \cos^3 2t \,dt$$

$\cos^2 2t$: $\frac{1+\cos 4t}{2}$; $\cos^3(2t)\cos 2t = (1-\sin^2 2t)\cos 2t$

Ex: refer Ch 8. ↓

$$= \frac{3}{8}\left[t - \frac{\sin 2t}{2} - \frac{1}{2}\left(t + \frac{\sin(4t)}{4}\right) + \frac{1}{2}\sin 2t - \frac{1}{3}\sin^3(2t) \right]_0^{\pi/2}$$

$$= \frac{3\pi}{32}$$

$$\text{Area} = 4A = \frac{3\pi}{8}$$

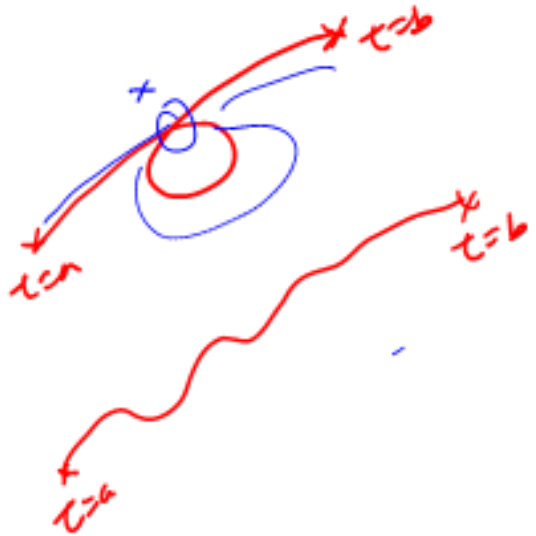
Length of a Parametrically Defined Curve

A curve $\mathcal{C}$, given by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, is a smooth curve if f and g are continuously differentiable on the interval $[a, b]$ with f' and g' are not simultaneously zero.

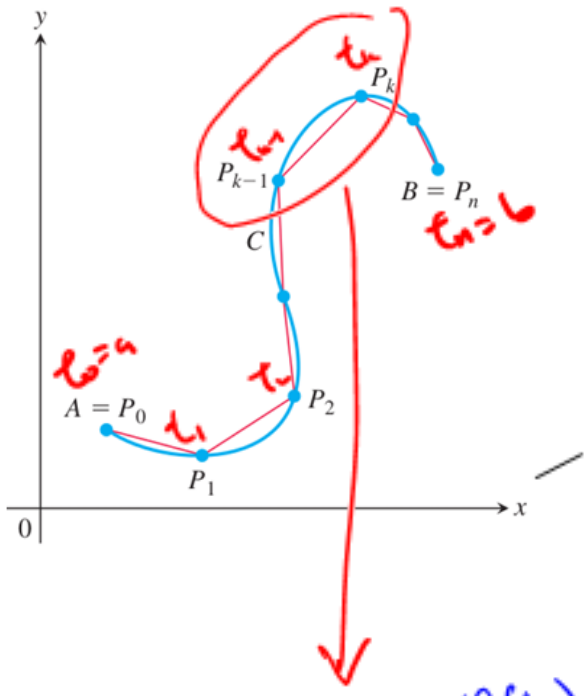
Length of a smooth curve

Let $\mathcal{C}$ be a smooth curve given by the equations $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$. If $\mathcal{C}$ is traversed exactly once as t increases from $t = a$ to $t = b$, then the length of $\mathcal{C}$ is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\ dt.$$



Length of a Parametrically Defined Curve



Subdivide the curve/path AB into

$P_0, P_1, P_2, \ldots, P_n$

A → B

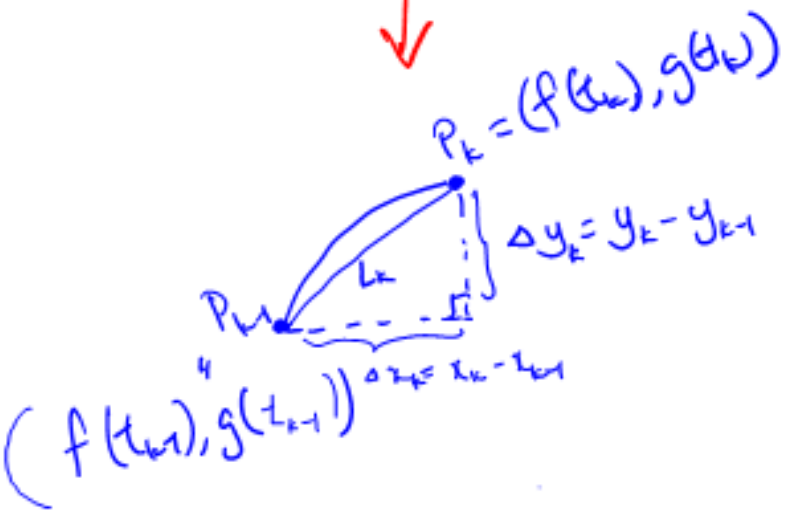
These points corresponding to a partition of the parameter interval $[a, b]$ by

$t_0 = a < t_1 < t_2 < \cdots < t_n = b$ where

$P_k = (f(t_k), g(t_k))$

$$L_k = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$$

$$= \sqrt{[f(t_k) - f(t_{k-1})]^2 + [g(t_k) - g(t_{k-1})]^2}$$



From the Mean Value Theorem, there is a c_1 in (t_{k-1}, t_k)

such that $$f'(c_1) = \frac{f(t_k) - f(t_{k-1})}{t_k - t_{k-1}}$$

$$\Rightarrow f(t_k) - f(t_{k-1}) = f'(c_1)(t_k - t_{k-1}) = f'(c_1)\,\Delta t_k$$

Similarly, from MVT, there is a c_2 in (t_{k-1}, t_k)

such that $$g'(c_2) = \frac{g(t_k) - g(t_{k-1})}{t_k - t_{k-1}}$$

$$\Rightarrow \quad g(t_k) - g(t_{k-1}) = g'(c_2)\,\Delta t_k$$

So, $$L_k = \int \sqrt{[f'(c_1)\Delta t_k]^2 + [g'(c_2)\Delta t_k]^2}$$

$$= \int \sqrt{(f'(c_1))^2 + (g'(c_2))^2}\;(\Delta t_k)$$

Sum of all $L_k = \sum_{k=1}^{n} L_k = \sum_{k=1}^{n} \sqrt{(f'(c_1))^2 + (g'(c_2))^2}\;(\Delta t_k)$

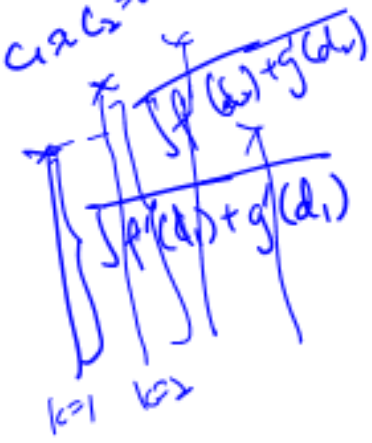
When we subdividing the path AB infinitely many numbers of times.

$$\sum_{k=1}^{\infty} L_k \approx AB.$$

So the length of curve $AB = \lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{(f'(c_1))^2 + (g'(c_2))^2}\,(\Delta t_k)$

when $n\to\infty$, $k=1$

$c_1, c_2 \in (t_0, t_k)$, $c_1 - c_2 = d_1$, $t_{k-1} - t_0 \to 0$ very similar to Riemann Sum.



$$= \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2}\,dt.$$

$x = f(t)$, $y = g(t)$

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt.$$

Length of a Parametrically Defined Curve

Example: Find the length of the circle of radius r defined parametrically by

treat r as a constant

$$x = r\cos t \quad \text{and} \quad y = r\sin t, \quad 0 \le t \le 2\pi.$$

As t varies from 0 to 2π, the circle 

traversed exactly once.

So, $$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{2\pi} \sqrt{(-r\sin t)^2 + (r\cos t)^2}\, dt$$

$$= \int_0^{2\pi} \sqrt{r^2 \sin^2 t + r^2 \cos^2 t}\, dt$$

$|r| = r$ because r is nonnegative.

$$= \int_0^{2\pi} r\, dt$$

$$= [rt]_0^{2\pi}$$

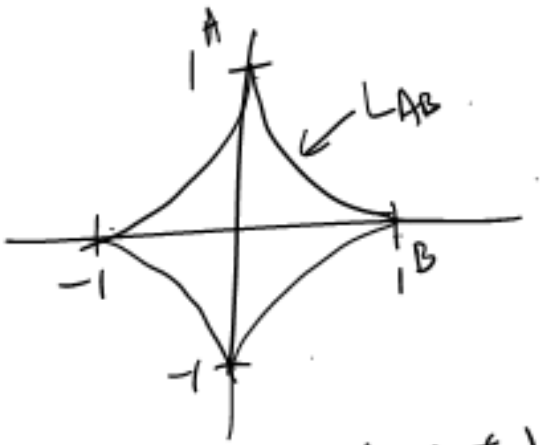
$$= r(2\pi - 0) = 2\pi r$$

(circumference of a circle with radius r)

Length of a Parametrically Defined Curve

Example: Find the length of the astroid

$$x = \cos^3 t, \quad y = \sin^3 t, \quad 0 \le t \le 2\pi.$$



By the symmetric of the curve

length $= 4L_{AB}$

$$L_{AB} = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\frac{dx}{dt} = 3\cos^2 t(-\sin t)$$

$$= -3\cos^2 t \sin t$$

$$\frac{dy}{dt} = 3\sin^2 t(\cos t)$$

$$= 3\sin^2 t \cos t$$

$$= \int_0^{\pi/2} \sqrt{\left(-3\cos^2 t \sin t\right)^2 + \left(3\sin^2 t \cos t\right)^2}\, dt$$

$$= \int_0^{\pi/2} \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t}\, dt$$

$$= \int_0^{\pi/2} \sqrt{9\cos^2 t \sin^2 t \left(\underbrace{\cos^2 t + \sin^2 t}_{=1}\right)}\, dt$$

$|3\cos t \sin t| = 3\cos t \sin t$ for $0 \le t \le \pi/2$

$$= \int_0^{\pi/2} 3\cos t \sin t \, dt$$

$\sin 2t = 2\cos t \sin t$

$$= \frac{3}{2} \int_0^{\pi/2} 2\cos t \sin t \, dt$$

$$= \frac{3}{2} \int_0^{\pi/2} \sin 2t \, dt$$

$$= \frac{3}{2} \left[\frac{-\cos 2t}{2} \right]_0^{\pi/2} = \frac{3}{2}$$

$$\text{length} = 4L_{AB} = 4\left(\frac{3}{2}\right) = 6$$